Solution to problem 6 on practice problems for exam 2

$\bar{Y}_{A.} = 61.3$

$\bar{Y}_{B.} = 38.67$

$\bar{Y}_{C.} = 72.33$

$\bar{Y}_{D.} = 86.67$

a) $H_0: \mu_A = \mu_B = \mu_C = \mu_D$

H_a: at least one of the population means is different from the others

From ANOVA table $p = 0.058 < 0.1 \Rightarrow$ reject H_0

Conclude that at least one of the population means is different from others.

b) $l = \frac{\mu_A + \mu_B}{2} - \frac{\mu_C + \mu_D}{2} \rightarrow \mu_A + \mu_B - \mu_C - \mu_D$

$\sum_{i=1}^{4} a_i = 1 + 1 - 1 - 1 = 0$ ✓

It is a linear contrast.

$a_1 = 1 \quad a_2 = 1 \quad a_3 = -1 \quad a_4 = -1$

c) $F = \frac{SSC}{MSE} = \frac{SSC}{323.33}$

$SSC = \frac{(\hat{l})^2}{\sum \frac{a_i^2}{n_i}} = \frac{n \hat{l}^2}{\sum a_i^2}$

$\sum a_i^2 = 4 \cdot 1^2 = 4$

$\hat{l} = \sum_{i=1}^{4} a_i \bar{Y}_{i.} = 1 \cdot 61.3 + 1 \cdot 38.67 - 72.33 - 86.67$

$= -59.03$

$SSC = \dfrac{3 \cdot (-59.03)^2}{4} = 2613.406$

$F = \dfrac{2613.403}{323.33} = 8.08$

$F_{0.05,1,8} = 5.32 \qquad (F_{0.1,1,8} = 3.46)$

Since $8.08 > 5.32$ we reject H_0 and conclude that there is a difference between the attendence rate for school A, B and school C, D.

d)

	B = 1	A = 2	C = 3	D = 4
$\bar{Y}_{i.}$	38.67	61.3	72.33	86.67

$\bar{y}_{4.} - \bar{y}_{1.} = 48 > LSD$

$\bar{y}_{3.} - \bar{y}_{1.} = 33.66 > LSD$

$\bar{y}_{2.} - \bar{y}_{1.} = 22.63 < LSD$

$\bar{y}_{4.} - \bar{y}_{2.} = 25.37 < LSD$

$\bar{y}_{4.} - \bar{y}_{3.} = 14.34$

$\alpha = 0.05$

$LSD = t_{\alpha/2,8} \sqrt{\dfrac{2MSE}{3}}$

$= 1.86 \sqrt{\dfrac{2 \cdot 323.33}{3}}$

$= \underline{\underline{27.3}}$

1 = B 2 = A 3 = C 4 = D

these 2 lines can be combined

e)

Contrast : $l = \mu_1 + \mu_2 - \mu_3 - \mu_4$

$$\hat{l} = -59.03$$

$$\hat{V}(\hat{l}) = MSE \cdot \Sigma \frac{a_i^2}{n}$$

$$= 323.33 \cdot \left(\frac{4}{3}\right) = 431.11$$

$$S = \sqrt{\hat{V}(\hat{l})} \sqrt{(t-1) F_{\alpha, df_1, df_2}}$$

$$F_{\alpha, df_1, df_2} = F_{0.05, 3, 8} = 4.07$$

$$S = \sqrt{431.11} \cdot \sqrt{3 \cdot 4.07} = 72.55$$

Since $|-59.03| = 59.03 < 72.55$ there is no significant difference between (School A & B) and (School C & D).